CHAPTER 2

Matrix Algebra

矩阵代数

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Chapter 2 Matrix Algebra

- § 2.1 Matrix Operation
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§ 2.1 Matrix Operation

- 1. Sums and Scalar Multiples
- 2. Matrix Multiplication
- 3. Properties of Matrix Multiplication
- 4. Powers of a Matrix
- 5. The Transpose of a Matrix

重点：熟悉矩阵乘法

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§ 2.1 Matrix Operation

A matrix is a rectangular array of numbers enclosed in square brackets. If there are m rows and n columns in the array, the matrix is called m x n and said have size m x n.

$$\begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}$$

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§ 2.1 Matrix Operation

- **Matrix Notation**

Two ways to denote $m \times n$ matrix A:

In terms of the *columns* of A:

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$$

In terms of the *entries* of A:

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}$$

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An n-dimensional vector is $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ just an n x 1 matrix. Such a matrix is called a column matrix.

A 1 x n matrix is called a row matrix because such a matrix looks like this $[\ a_1 \quad a_2 \quad \cdots \quad a_n\ ]$:

Main diagonal entries: $a_{11}, a_{22}, a_{33}, \cdots$ 主对角线元素

Diagonal Matrix(对角矩阵)： A square matrix, and nondiagonal entries $\equiv 0$

$$\begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$$

For any positive integers m and n, there is an m x n zero matrix （零矩阵）, which we denote with boldface type 0.

Zero matrix:

$$\mathbf{0} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{bmatrix}$$

§ 2.1 Matrix Operation

- **Sums and Scalar Multiples**

  A, B : $m \times n$ matrices

  A + B : $m \times n$ matrices,

  each entry in A + B is the sum of the corresponding entries in A and B.

The sum A + B is defined only when A and B are the same size.

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Example :

$$A=\begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix},\ B=\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix},\ C=\begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix}$$

Then

$$A+B=\begin{bmatrix} 5 & 1 & 6 \\ 2 & 8 & 9 \end{bmatrix}$$

$$2B=2\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix}=\begin{bmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{bmatrix}$$

$$A-2B=\begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix}-\begin{bmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{bmatrix}=\begin{bmatrix} 2 & -2 & 3 \\ -7 & -7 & -12 \end{bmatrix}$$

But A + C is not defined because A and C have different sizes.

§ 2.1 Matrix Operation

THEOREM 1

Let A, B, and C be matrices of the same size, and let r and s be scalars. Then

a. $A + B = B + A$

b. $(A + B) + C = A + (B + C)$

c. $A + 0 = A$

d. $r(A + B) = rA + rB$

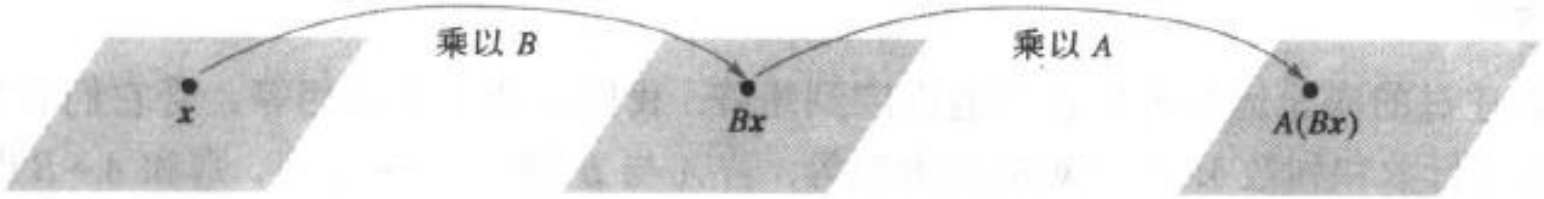
e. $(r + s)A = rA + sA$

f. $r(sA) = (rs)A$

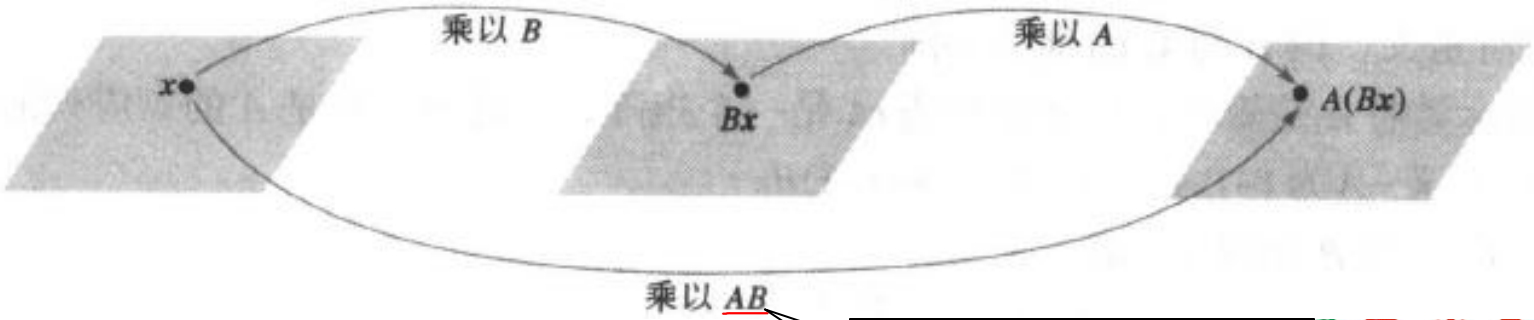
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§ 2.1 Matrix Operation

- **Matrix Multiplication**



$$A(Bx) = (AB)x$$



A single matrix

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Suppose A is $m \times n$ and B is $n \times p$ where

$$B = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p \end{bmatrix} \text{ and } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}.$$

Then

$$B\mathbf{x} = x_1\mathbf{b}_1 + x_2\mathbf{b}_2 + \cdots + x_p\mathbf{b}_p$$

and

$$A(B\mathbf{x}) = A(x_1\mathbf{b}_1 + x_2\mathbf{b}_2 + \cdots + x_p\mathbf{b}_p)$$

$$= A(x_1\mathbf{b}_1) + A(x_2\mathbf{b}_2) + \cdots + A(x_p\mathbf{b}_p)$$

$$= x_1A\mathbf{b}_1 + x_2A\mathbf{b}_2 + \cdots + x_pA\mathbf{b}_p = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}.$$

The vector A(Bx) is a linear combination of the vectors $Ab_1,\ldots,Ab_p$, using the entries in x as weights.

Therefore,

$$A(B\mathbf{x}) = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \end{bmatrix}\mathbf{x}.$$

and by defining

$$\boxed{AB = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \end{bmatrix}}$$

we have $A(B\mathbf{x}) = (AB)\mathbf{x}$.

§ 2.1 Matrix Operation

Definition

If A is an $m \times n$ matrix, and if B is an $n \times p$ matrix with columns $b_1,\ldots,b_p$, then the product AB is the $m \times p$ matrix whose columns are $Ab_1,\ldots, Ab_p$.

That is

$$AB = A[b_1 \quad b_2 \quad \cdots \quad b_p] = [Ab_1 \quad Ab_2 \quad \cdots \quad Ab_p]$$

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- Example：Compute AB, where $A=\begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}$, $B=\begin{bmatrix}4 & 3 & 6\\ 1 & -2 & 3\end{bmatrix}$

Sol： Write $B=[\boldsymbol{b}_1\ \ \boldsymbol{b}_2\ \ \boldsymbol{b}_3]$, and compute:

$$A\boldsymbol{b}_1=\begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}\begin{bmatrix}4\\ 1\end{bmatrix}=\begin{bmatrix}11\\ -1\end{bmatrix},\quad A\boldsymbol{b}_2=\begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}\begin{bmatrix}3\\ -2\end{bmatrix}=\begin{bmatrix}0\\ 13\end{bmatrix},\quad A\boldsymbol{b}_3=\begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}\begin{bmatrix}6\\ 3\end{bmatrix}=\begin{bmatrix}21\\ -9\end{bmatrix}$$

$$AB=A[\boldsymbol{b}_1\ \ \boldsymbol{b}_2\ \ \boldsymbol{b}_3]=\begin{bmatrix}11 & 0 & 21\\ -1 & 13 & -9\end{bmatrix}$$

$A\boldsymbol{b}_1\ A\boldsymbol{b}_2\ A\boldsymbol{b}_3$

Each column of AB is a linear combination of the columns of A using weights from the corresponding column of B.

EXAMPLE: If A is 4×3 and B is 3×2, then what are the sizes of AB and BA?

Solution:

$$AB = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix} = \begin{bmatrix} * & * \\ * & * \\ * & * \\ * & * \end{bmatrix}$$

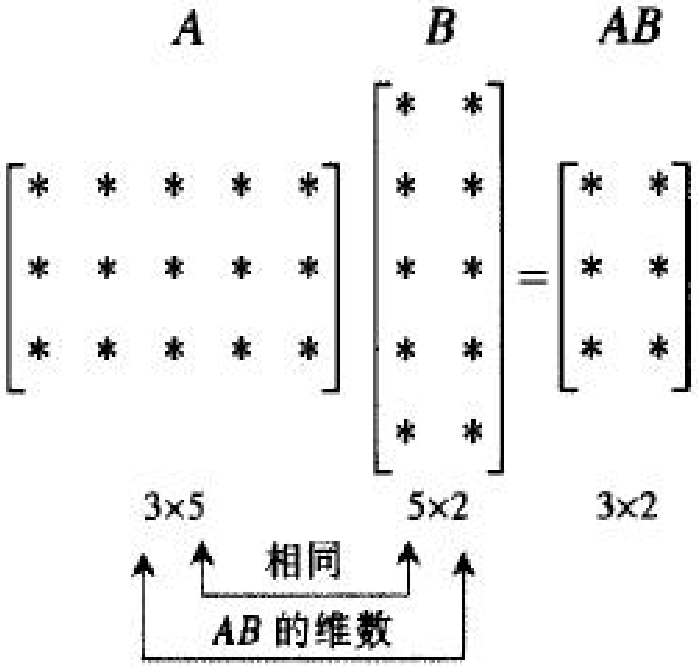
BA would be $\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix} \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}$

which is undefined.

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$.

- Example ： If A is 3×5 matrix and B is a 5×2 matrix, what are the sizes of AB and BA, if they are defined?

  Sol：(1)Since A has 5 columns and B has 5 rows, the product AB is defined and is a 3×2 matrix:



(2)∵ the 2 columns of B do not match the 3 row of A. ∴The product BA is not defined

§ 2.1 Matrix Operation

Row- Column Rule for Computing AB

If the product AB is defined, then the entry in row i and column j of AB is the sum of the product of corresponding entries from row i of A and column j of B. If $(AB)_{ij}$ denotes the (i, j)-entry in AB, and if A is an $m \times n$ matrix, then $(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$

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The definition

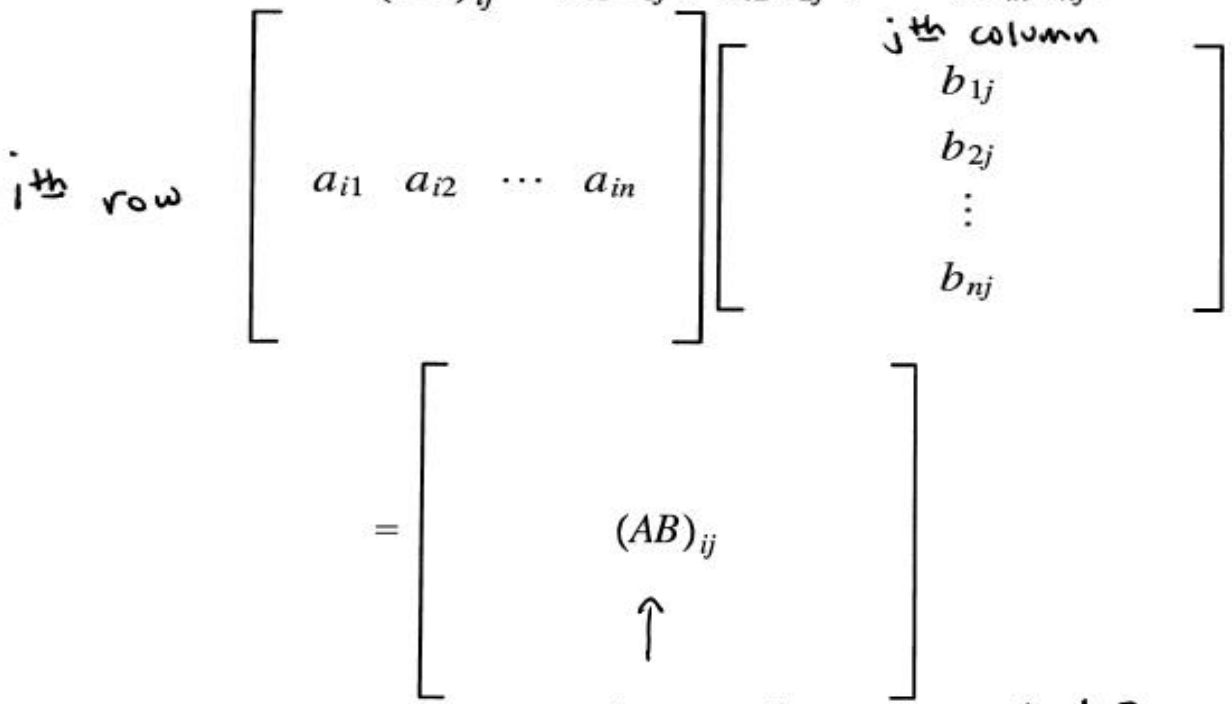
$$AB = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \end{bmatrix}$$

is good for theoretical work.

When A and B have small sizes, the following method is more efficient when working by hand.

If AB is defined, let $(AB)_{ij}$ denote the entry in the ith row and jth column of AB. Then

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$



EXAMPLE $A = \begin{bmatrix} 2 & 3 & 6 \\ -1 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -3 \\ 0 & 1 \\ 4 & -7 \end{bmatrix}$. Compute AB, if it is defined.

Solution: Since A is 2×3 and B is 3×2, then AB is defined and AB is $\underline{2} \times \underline{2}$.

$$AB = \begin{bmatrix} \mathbf{2} & \mathbf{3} & \mathbf{6} \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{2} & -3 \\ \mathbf{0} & 1 \\ \mathbf{4} & -7 \end{bmatrix} = \begin{bmatrix} \mathbf{28} & \blacksquare \\ \blacksquare & \blacksquare \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{2} & \mathbf{3} & \mathbf{6} \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & \mathbf{-3} \\ 0 & \mathbf{1} \\ 4 & \mathbf{-7} \end{bmatrix} = \begin{bmatrix} 28 & \mathbf{-45} \\ \blacksquare & \blacksquare \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 6 \\ \mathbf{-1} & \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} \mathbf{2} & -3 \\ \mathbf{0} & 1 \\ \mathbf{4} & -7 \end{bmatrix} = \begin{bmatrix} 28 & -45 \\ \mathbf{2} & \blacksquare \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 6 \\ \mathbf{-1} & \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} 2 & \mathbf{-3} \\ 0 & \mathbf{1} \\ 4 & \mathbf{-7} \end{bmatrix} = \begin{bmatrix} 28 & -45 \\ 2 & \mathbf{-4} \end{bmatrix}$$

So $AB = \begin{bmatrix} 28 & -45 \\ 2 & -4 \end{bmatrix}$.

- Example : Find the entries in the second row of AB, where

$$A=\begin{bmatrix} 2 & -5 & 0 \\ -1 & 3 & -4 \\ 6 & -8 & -7 \\ -3 & 0 & 9 \end{bmatrix},\quad B=\begin{bmatrix} 4 & -6 \\ 7 & 1 \\ 3 & 2 \end{bmatrix}$$

Sol:

$$\rightarrow\begin{bmatrix} 2 & -5 & 0 \\ -1 & 3 & -4 \\ 6 & -8 & -7 \\ -3 & 0 & 9 \end{bmatrix}\begin{bmatrix} 4 & -6 \\ 7 & 1 \\ 3 & 2 \end{bmatrix}=\begin{bmatrix} \square & \square \\ -4+21-12 & 6+3-8 \\ \square & \square \\ \square & \square \end{bmatrix}=\begin{bmatrix} \square & \square \\ 5 & 1 \\ \square & \square \\ \square & \square \end{bmatrix}$$

$$\text{row}_i(AB)=\text{row}_i(A)\cdot B$$

$$\begin{bmatrix} -1 & 3 & -4 \end{bmatrix}\begin{bmatrix} 4 & -6 \\ 7 & 1 \\ 3 & 2 \end{bmatrix}=\begin{bmatrix} 5 & 1 \end{bmatrix}$$

Properties of Matrix Multiplication

Theorem 2

Let A be an m×n matrix, and let B and C have sizes for which the indicated sums and products are defined.

a. $A(BC) = (AB)C$ （乘法结合律）

b. $A(B+C) = AB + AC$ （乘法左分配律）

c. $(B+C)A = BA + CA$ （乘法右分配律）

d. $r(AB) = (rA)B = A(rB)$， r 为任意数

e. $I_m A = A = AI_n$ （矩阵乘法的恒等式）

Properties above are analogous to properties of real numbers. But **NOT ALL** real number properties correspond to matrix properties.

- Example :

Let $A=\begin{bmatrix}5 & 1\\ 3 & -2\end{bmatrix}$ $B=\begin{bmatrix}2 & 0\\ 4 & 3\end{bmatrix}$

Show that these matrices do not commute.

That is, verify that $AB \neq BA$

Solution:

$$AB=\begin{bmatrix}5 & 1\\ 3 & -2\end{bmatrix}\begin{bmatrix}2 & 0\\ 4 & 3\end{bmatrix}=\begin{bmatrix}14 & 3\\ -2 & -6\end{bmatrix}$$

$$BA=\begin{bmatrix}2 & 0\\ 4 & 3\end{bmatrix}\begin{bmatrix}5 & 1\\ 3 & -2\end{bmatrix}=\begin{bmatrix}10 & 2\\ 29 & -2\end{bmatrix}$$

- Example :

Let $A=\begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}, B=\begin{bmatrix} 8 & 4 \\ 5 & 5 \end{bmatrix}, C=\begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$

verify that AB = AC, but B !=C

Solution:

$$AB=\begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}\begin{bmatrix} 8 & 4 \\ 5 & 5 \end{bmatrix}=\begin{bmatrix} 1 & -7 \\ -2 & 14 \end{bmatrix},$$

$$AC=\begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}\begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}=\begin{bmatrix} 1 & -7 \\ -2 & 14 \end{bmatrix}$$

- Example :

Let $A=\begin{bmatrix} 3 & -6 \\ -1 & 2 \end{bmatrix}$, there is existing a 2 x 2 matrix which is not zero matrix, that makes AB = 0;

Solution:

Consider $B = [\mathbf{b}_1 \ \ \mathbf{b}_2]$. To make $AB = 0$, one needs $A\mathbf{b}_1 = \mathbf{0}$ and $A\mathbf{b}_2 = \mathbf{0}$. By inspection of A, a suitable $\mathbf{b}_1$ is $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, or any multiple of $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Example: $B = \begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix}$.

§ 2.1 Matrix Operation

- **Warnings:**

  1. In general, AB ≠ BA.

  2. The cancellation laws do not hold for matrix multiplication. That is, if AB =AC, then it is not true in general that B = C.

  3. If a product AB is the zero matrix, you cannot conclude in general that either A = 0 or B = 0.

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Power of a Matrix

If A is an $n \times n$ matrix and if k is a positive integer, then A^k denotes the product of k copies of A:

$$A^k = \underbrace{A \cdots A}_{k\text{个}}$$

(1) If A is nonzero and if $\mathbf{x}$ is in R^n, then $A^k\mathbf{x}$ is the result of left-multiplying $\mathbf{x}$ by repeatedly k times.

(2) If k = 0, then $A^0\mathbf{x}$ should be $\mathbf{x}$ itself.

EXAMPLE:

$$\begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}^3 = \left(\begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix} \right) \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 9 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 21 & 8 \end{bmatrix}$$

§ 2.1 Matrix Operation

The Transpose of a Matrix (转置操作)

Given an m×n matrix A, the transpose of A is the n×m matrix, denoted by A^T, whose columns are formed from the corresponding rows of A.

$$[A^T]_{ij} = [A]_{ji}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} -5 & 2 \\ 1 & -3 \\ 0 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -3 & 5 & -2 & 7 \end{bmatrix}$$

$$A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}, \quad B^T = \begin{bmatrix} -5 & 1 & 0 \\ 2 & -3 & 4 \end{bmatrix}, \quad C^T = \begin{bmatrix} 1 & -3 \\ 1 & 5 \\ 1 & -2 \\ 1 & 7 \end{bmatrix}$$

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EXAMPLE: Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ -2 & 4 \end{bmatrix}$. Compute AB, $(AB)^T$, A^TB^T and B^TA^T.

Solution:

$$AB = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 1 & 10 \end{bmatrix}$$

$$(AB)^T = \begin{bmatrix} 1 & 1 \\ 4 & 10 \end{bmatrix}$$

$$A^TB^T = \begin{bmatrix} 1 & 3 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 7 & 3 & 10 \\ 2 & 0 & -4 \\ 2 & 1 & 4 \end{bmatrix}$$

$$B^TA^T = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 4 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & 10 \end{bmatrix}$$

Theorem 3

Let A and B denote matrices, whose sizes are appropriate for the following sums and products.

a. $(A^T)^T = A$.

b. $(A+B)^T = A^T + B^T$.

c. 对任意数 r， $(rA)^T = rA^T$.

d. $(AB)^T = B^T A^T$. 思考题：证明(d)

EXAMPLE: Prove that $(ABC)^T = C^T B^T A^T$.

Solution: By Theorem 3d,

$$(ABC)^T = ((AB)C)^T = C^T(AB)^T$$

$$= C^T(B^T A^T) = C^T B^T A^T.$$

Assignment

- Ex2.1: 17，21，22，23

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§ 2.2 The Inverse of a Matrix

- 1. The Inverse of a Matrix
- 2. Elementary Matrices
- 3. An Algorithm for Finding A^{-1}
- 4. Another View of Matrix Inversion

重点：矩阵的逆的定义（类似于数的倒数），怎么求矩阵的逆

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§ 2.2 The Inverse of a Matrix

The Inverse of a Matrix（ 矩阵的逆 ）

Recall the multiplicative inverse of a number:

$$5^{-1} \cdot 5 = 1 \text{ 和 } 5 \cdot 5^{-1} = 1$$

For matrix: An n×n matrix A is said to be ***invertible*** （可逆的）, if there is an n×n matrix C such that CA = I and AC = I.

In this case, C is inverse of A. And C is unique. The unique inverse is denoted by A^{-1}, so that $A^{-1}A = I$ and $AA^{-1} = I$.

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FACT If A is invertible, then the inverse is unique.

Proof: Assume B and C are both inverses of A. Then

$$B = BI = B(\underline{AC}) = (\underline{BA})\underline{C} = I\underline{C} = C.$$

So the inverse is unique since any two inverses coincide. ■

The inverse of A is usually denoted by A^{-1}.

We have

$$\boxed{AA^{-1} = A^{-1}A = I_n}$$

Not all $n \times n$ ***matrices are invertible.*** A matrix which is *not* invertible is sometimes called a **singular** matrix. An invertible matrix is called **nonsingular** matrix.

Singular: 奇异的

§ 2.2 The Inverse of a Matrix

- **Example ：**

If $A=\begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}, C=\begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix},$

Then

$$AC=\begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}\begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$CA=\begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Thus $C = A^{-1}$

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§ 2.2 The Inverse of a Matrix

Theorem 4

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If $ad - bc = 0$, then A is not invertible.

参考习题2.2中的习题25,26

The quantity *ad-bc* is called the ***determinant*(行列式)** of A, denoted by

$$\det A = ad - bc$$

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Assume A is any invertible matrix and we wish to solve $A\mathbf{x} = \mathbf{b}$
Then

$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}$ and so

$I\mathbf{x} = A^{-1}\mathbf{b}$ or $\mathbf{x} = A^{-1}\mathbf{b}$.

Suppose $\mathbf{w}$ is also a solution to $A\mathbf{x} = \mathbf{b}$. Then $A\mathbf{w} = \mathbf{b}$ and

$A^{-1}A\mathbf{w} = A^{-1}\mathbf{b}$ which means $\mathbf{w} = A^{-1}\mathbf{b}$.

So, $\mathbf{w} = A^{-1}\mathbf{b}$, which is in fact the same solution.

We have proved the following result:

Theorem 5

If A is an invertible $n \times n$ matrix, then for each $\mathbf{b}$ in $\mathbf{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.

EXAMPLE: Use the inverse of $A = \begin{bmatrix} -7 & 3 \\ 5 & -2 \end{bmatrix}$ to solve

$$\begin{aligned} -7x_1 + 3x_2 &= 2 \\ 5x_1 - 2x_2 &= 1 \end{aligned}.$$

Solution: Matrix form of the linear system:

$$\underset{A}{\begin{bmatrix} -7 & 3 \\ 5 & -2 \end{bmatrix}} \underset{\underline{x}}{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} = \underset{b}{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}$$

$$A^{-1} = \frac{1}{14-15}\begin{bmatrix} -2 & -3 \\ -5 & -7 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}.$$

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 17 \end{bmatrix}$$

- Remark:

  Theorem 5 is seldom used to solve linear systems, since the previous method of row reduction is nearly always faster.

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Theorem 6 Suppose A and B are invertible. Then the following results hold:

a. A^{-1} is invertible and $(A^{-1})^{-1} = A$ (i.e. A is the inverse of A^{-1}).

b. AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$

c. A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$

Partial proof of part b:

$$(AB)(B^{-1}A^{-1}) = A(\underline{B\,B^{-1}})A^{-1}$$

$$= A(\underline{\quad I \quad})A^{-1} = \underline{A\,A^{-1}} = \underline{\quad I \quad}.$$

Similarly, one can show that $(B^{-1}A^{-1})(AB) = I$.

Theorem 6, part b can be generalized to three or more invertible matrices:

$$(ABC)^{-1} = \underline{C^{-1}B^{-1}A^{-1}}$$

§ 2.2 The Inverse of a Matrix

Earlier, we saw a formula for finding the inverse of a 2×2 invertible matrix. How do we find the inverse of an invertible $n \times n$ matrix? To answer this question, we first look at **elementary** matrices.

- **Elementary Matrices (初等矩阵)**

  An elementary matrix is one that is obtained by performing a **single** elementary row operation on an **identity matrix**.

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EXAMPLE: Let $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$,

$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$ and $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$.

E_1, E_2, and E_3 are elementary matrices. Why?

Observe the following products and describe how these products can be obtained by elementary row operations on A.

$$E_1A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ 2d & 2e & 2f \\ g & h & i \end{bmatrix}$$

$$E_2A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ g & h & i \\ d & e & f \end{bmatrix}$$

$$E_3A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 3a+g & 3b+h & 3c+i \end{bmatrix}$$

§ 2.2 The Inverse of a Matrix

If an elementary row operation is performed on an m×n matrix A, the resulting matrix can be written as EA, where the m×m matrix E is created by performing the same row operation on I_m.

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Elementary matrices are *invertible* because row operations are *reversible*. To determine the inverse of an elementary matrix E, determine the elementary row operation needed to transform E back into I and apply this operation to I to find the inverse.

For example,

$$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \qquad E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

Example: Let $A = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$. Then

$$E_1A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$E_2(E_1A) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

$$E_3(E_2E_1A) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So

$$\boxed{E_3E_2E_1A = I_3}.$$

Then multiplying on the right by A^{-1}, we get

$$E_3E_2E_1A\underline{\quad A^{-1}\quad} = I_3\underline{\quad A^{-1}\quad}.$$

So

$$\boxed{E_3E_2E_1I_3 = A^{-1}}$$

The elementary row operations that row reduce A to $\mathbf{I}_n$ are the same elementary row operations that transform $\mathbf{I}_n$ into $\mathbf{A}^{-1}$.

Theorem 7

A n×n matrix *A* is invertible if and only if *A* is row equivalent to I_n, and in this case, any sequence of elementary row operations that reduces *A* to I_n also transforms I_n into A^{-1}

Proof: **(i) Suppose that A is invertible.** Then, since the equation Ax = b has a solution for each b, by Th5 A has a pivot position in every row. Because A is square, the n pivot positions must be on the diagonal, which implies that $\mathbf{A \sim I_n}$.

(ii) Now suppose, conversely, that $A \sim I_n$.

Then, there exist elementary matrices $E_1,\ldots, E_p$ such that

$$\mathbf{E_p \ldots E_1 A = I_n}$$

Since the product $E_p E_{p-1} \ldots E_1$ is invertible, we have

$$(E_p \ldots E_1)^{-1} (E_p \ldots E_1) A = (E_p \ldots E_1)^{-1} I_n$$

That is $A = (E_p E_{p-1} \ldots E_1)^{-1}$

Thus A is invertible, as it is the inverse of an invertible matrix.

Furthermore, we have

$$A^{-1} = \left[(E_p \ldots E_1)^{-1} \right]^{-1} = E_p \ldots E_1 = E_p \ldots E_1 I$$

§ 2.2 The Inverse of a Matrix

- **An Algorithm for Finding A^{-1}**

Algorithm for Finding A^{-1}

Row reduce the augmented matrix [A I]. If A is row equivalent to I, then [A I] is row equivalent to [I A^{-1}]. Otherwise, A does not have an inverse.

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- Example : Find the inverse of the matrix A, if it exists.

$$A=\begin{bmatrix}0 & 1 & 2\\ 1 & 0 & 3\\ 4 & -3 & 8\end{bmatrix}$$

Sol:

$$[A\ \ I]=\left[\begin{array}{ccc|ccc}0 & 1 & 2 & 1 & 0 & 0\\ 1 & 0 & 3 & 0 & 1 & 0\\ 4 & -3 & 8 & 0 & 0 & 1\end{array}\right]\sim\left[\begin{array}{ccc|ccc}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 4 & -3 & 8 & 0 & 0 & 1\end{array}\right]$$

$$\sim\left[\begin{array}{ccc|ccc}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 0 & -3 & -4 & 0 & -4 & 1\end{array}\right]\sim\left[\begin{array}{ccc|ccc}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 0 & 0 & 2 & 3 & -4 & 1\end{array}\right]$$

$$\sim\left[\begin{array}{ccc|ccc}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 0 & 0 & 1 & 3/2 & -2 & 1/2\end{array}\right]\sim\left[\begin{array}{ccc|ccc}1 & 0 & 0 & -9/2 & 7 & -3/2\\ 0 & 1 & 0 & -2 & 4 & -1\\ 0 & 0 & 1 & 3/2 & -2 & 1/2\end{array}\right]$$

Sol:

Since A ~ *I*, we conclude that A is invertible by Theorem 7 and

$$A^{-1} = \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix}$$

To check the final answer:

$$AA^{-1} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix} \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

EXAMPLE: Find the inverse of $A = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, if it exists.

Solution:

$$[A \;\; I] = \begin{bmatrix} 2 & 0 & 0 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & \frac{3}{2} & 1 & 0 \end{bmatrix}$$

(A | I)

So $A^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 \\ \frac{3}{2} & 1 & 0 \end{bmatrix}$

Assignment

- Ex2.2：21，31，32

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补充

初等行（列）变换

定理　对一个$m \times n$ 矩阵 A作一初等行变换就相当于在A 的左边乘上相应的$m \times m$ 初等矩阵；对A作一初等列变换就相当于在A 的右边乘上相应的$n \times n$ 初等矩阵。

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思考题

- 矩阵A 称为对称矩阵，如果$A = A^T$;矩阵A 称为反对称矩阵，如果$A = -A^T$。
- 证明：任一n × n 矩阵都可以表为一对称矩阵与一反对称矩阵之和。

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§ 2.3 Characterizations of Invertible Matrices

Theorem 8 The Invertible Matrix Theorem

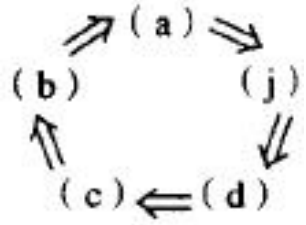
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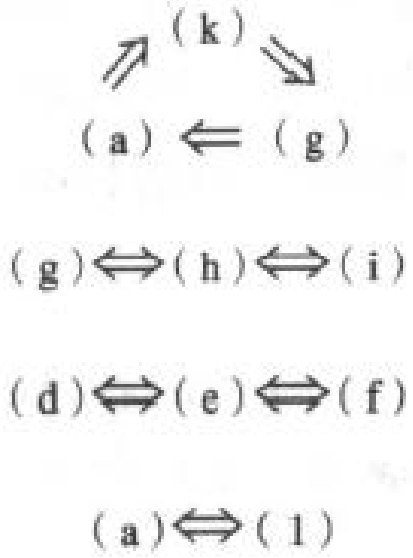
Th8: Let A be a n*n square matrix. Then the following statements are equivalent.

a. A is an invertible matrix.
b. A is row equivalent to the identity matrix $\mathbf{I_n}$.
c. A has n pivot positions.
d. The equation Ax = 0 has only the trivial solution.
e. The columns of A form a linearly independent set.
f. The linear transformation $x \rightarrow Ax$ is one-to-one.
g. The equation Ax = b has at least one solution for each b in R^n.
h. The columns of A span R^n.
i. The linear transformation $x \rightarrow Ax$ maps R^n onto R^n.
j. There is an $n \times n$ matrix C such that CA = I.
k. There is an $n \times n$ matrix D such that AD = I.
l. A^T is an invertible matrix.

§ 2.3 Characterizations of Invertible Matrices

- The “circle” of Implication





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Remark:

1.Theorem 8 applies only to square matrix

2. It provides the relation among many important concepts

- Example : Use the Invertible Matrix Theorem to decide if A is invertible:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix}$$

Solution:

$$A \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

So A has 3 pivot positions and hence is invertible. (By Theorem 8c)

§ 2.3 Characterizations of Invertible Matrices

Invertible Linear Transformations

A linear transformation T: R^n->R^n is said to be invertible if there exists a transformation S: R^n->R^n such that

$$S(T(\mathbf{x})) = \mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } R^n \qquad (1)$$

$$T(S(\mathbf{x})) = \mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } R^n \qquad (2)$$

S is T's invertible , $S=T^{-1}$.



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§ 2.3 Characterizations of Invertible Matrices

Theorem 9

Let T: R^n->R^n be a linear transformation and let A be the standard matrix for T. Then T is invertible if and only if A is an invertible matrix. In that case, the linear transformation S given by $S(x) = A^{-1}x$ is the unique function satisfying (1) and (2).

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- Proof:

Suppose that T is invertible. Then (2) shows that T is onto R^n, for if b is in R^n and x = S(b), then T(x) = T(S(b)) = b, so each b is in the range of T. Thus, by theorem 12 in section 1.9, the columns of A span R^n. Furthermore, A is invertible, by the invertible Matrix Theorem, statement(h).

Conversely, suppose and A is invertible, and let $S(x) = A^{-1}x$. There holds that for all $x \in R^n$,

$$S\big(T(x)\big) = S(Ax) = A^{-1}(Ax) = x$$

$$T(S(x)) = T(A^{-1}x) = A((A^{-1}x)) = x$$

Thus T is invertible. The proof that S is unique is outlined in Exercise 39.

Assingment

- Ex2.3: 27, 28，33

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§ 2.4 Partitioned Matrices

- 1. Partitions of Matrices(分块矩阵)
- 2. Addition and Scalar Multiplication
- 3. Multiplication of Partitioned Matrices
- 4. Inverses of Partitioned Matrices

重点掌握分块矩阵的乘法法则

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§ 2.4 Partitioned Matrices

Partitions of Matrices

Example： The matrix

$$A=\left[\begin{array}{ccc|cc|c} 3 & 0 & -1 & 5 & 9 & -2 \\ -5 & 2 & 4 & 0 & -3 & 1 \\ \hline -8 & -6 & 3 & 1 & 7 & -4 \end{array}\right]$$

2 × 3 partitioned matrix

$$A=\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

where

$$A_{11}=\begin{bmatrix} 3 & 0 & -1 \\ -5 & 2 & 4 \end{bmatrix},\quad A_{12}=\begin{bmatrix} 5 & 9 \\ 0 & -3 \end{bmatrix},\quad A_{13}=\begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$A_{21}=[-8 \quad -6 \quad 3],\quad A_{22}=[1 \quad 7],\quad A_{23}=[-4]$$

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§ 2.4 Partitioned Matrices

- **Addition and Scalar Multiplication**

A and B : Matrices of same size and partitioned in the same way

A + B: the same partition of the ordinary matrix sum A + B.
each block is the sum of corresponding blocks of A and B.

*c*A: Multiplication of a partitioned matrix A by a scalar *c* computed block by block.

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§ 2.4 Partitioned Matrices

- **Multiplication of Partitioned Matrices: AB**
  - by the usual row – column rule
  - the column partition A matches the row partition of B
- Example ： Let

$$A=\left[\begin{array}{ccc|cc} 2 & -3 & 1 & 0 & -4 \\ 1 & 5 & -2 & 3 & -1 \\ \hline 0 & -4 & -2 & 7 & -1 \end{array}\right]=\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, B=\left[\begin{array}{cc} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ \hline -1 & 3 \\ 5 & 2 \end{array}\right]=\begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

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Solution:

where

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} A_{11}B_1 + A_{12}B_2 \\ A_{21}B_1 + A_{22}B_2 \end{bmatrix} = \left[\begin{array}{cc} -5 & 4 \\ -6 & 2 \\ \hline 2 & 1 \end{array}\right]$$

$$A_{11}B_1 = \begin{bmatrix} 2 & -3 & 1 \\ 1 & 5 & -2 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix}$$

$$A_{12}B_2 = \begin{bmatrix} 0 & -4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix}$$

$$A_{11}B_1 + A_{12}B_2 = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix} + \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ -6 & 2 \end{bmatrix}$$

- Example： Let $A=\begin{bmatrix}-3 & 1 & 2\\ 1 & -4 & 5\end{bmatrix}$ and $B=\begin{bmatrix}a & b\\ c & d\\ e & f\end{bmatrix}$.

Verify that

$$AB=\mathrm{col}_1(A)\mathrm{row}_1(B)+\mathrm{col}_2(A)\mathrm{row}_2(B)+\mathrm{col}_3(A)\mathrm{row}_3(B)$$

Solution： Each term above is an ***outer product*** **(外积)**

By row-column rule:

$$\mathrm{col}_1(A)\mathrm{row}_1(B)=\begin{bmatrix}-3\\ 1\end{bmatrix}\begin{bmatrix}a & b\end{bmatrix}=\begin{bmatrix}-3a & -3b\\ a & b\end{bmatrix}$$

$$\mathrm{col}_2(A)\mathrm{row}_2(B)=\begin{bmatrix}1\\ -4\end{bmatrix}\begin{bmatrix}c & d\end{bmatrix}=\begin{bmatrix}c & d\\ -4c & -4d\end{bmatrix}$$

$$\mathrm{col}_3(A)\mathrm{row}_3(B)=\begin{bmatrix}2\\ 5\end{bmatrix}\begin{bmatrix}e & f\end{bmatrix}=\begin{bmatrix}2e & 2f\\ 5e & 5f\end{bmatrix}$$

$$\sum_{k=1}^{3}\mathrm{col}_k(A)\mathrm{row}_k(B)=\begin{bmatrix}-3a+c+2e & -3b+d+2f\\ a-4c+5e & b-4d+5f\end{bmatrix}$$

§ 2.4 Partitioned Matrices

Theorem 10： Column-Row Expansion of AB

If A is m×n matrix and B is n×p, then

$$AB = [\mathrm{col}_1(A) \quad \mathrm{col}_2(A) \quad \cdots \quad \mathrm{col}_n(A)] \begin{bmatrix} \mathrm{row}_1(B) \\ \mathrm{row}_2(B) \\ \vdots \\ \mathrm{row}_n(B) \end{bmatrix}$$

$$= \mathrm{col}_1(A)\mathrm{row}_1(B) + \cdots + \mathrm{col}_n(A)\mathrm{row}_n(B)$$

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Proof:

For each row index i and columns index j, the (i,j)-entry in $col_k(A)row_k(B)$ is the product of a_{ik} from $col_k(A)$ and b_{kj} from $row_k(B)$.

Hence the (i,j)- entry in the sum shown in Theorem10 is

$$a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{in}b_{nj}$$

This sum is also the (i,j)- entry in AB, by the row-column rule.

§ 2.4 Partitioned Matrices

Inverses of Partitioned Matrices

Example： ***Block upper triangular***

$$A=\begin{bmatrix} A_{11} & A_{22} \\ 0 & A_{22} \end{bmatrix}$$

Assume that A_{11} is $p\times p$, A_{22} is $q\times q$, and A is invertible.

Find a formula for A^{-1}.

Solution： $B = A^{-1}$

$$\begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}=\begin{bmatrix} I_p & 0 \\ 0 & I_q \end{bmatrix} \qquad (2)$$

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That is,

$$A_{11}B_{11} + A_{12}B_{21} = I_p \quad (3)$$

$$A_{11}B_{12} + A_{12}B_{22} = 0 \quad (4)$$

$$A_{22}B_{21} = 0 \quad (5)$$

$$A_{22}B_{22} = I_q \quad (6)$$

∵ Using the Invertible Matrix Theorem and A_{22} is square

∴ A_{22} is invertible and $B_{22} = A_{22}^{-1}$

Then, Using (5)and find $B_{21} = A_{22}^{-1}0 = 0$

so that (3) simplifies to $A_{11}B_{11} + 0 = I_p$

Finally, from (4) $A_{11}B_{12} = -A_{12}B_{22} = -A_{12}A_{22}^{-1}$ 和 $B_{12} = -A_{11}^{-1}A_{12}A_{22}^{-1}$

- Thus

$$A^{-1} = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}^{-1} = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1} A_{12} A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}$$

A block diagonal matrix is a partitioned matrix with zero blocks off the main diagonal. Such a matrix is invertible if and only if each block on the diagonal is invertible.

分块乘法的初等变换及应用举例

- 将分块乘法与初等变换结合就成为矩阵运算中极端重要的手段
- 现将某个单位矩阵如下进行分块：
- $\begin{bmatrix} I_m & O \\ O & I_n \end{bmatrix}$
- 对它进行两行（列）对换；某一行（列）左乘（右乘）一个矩阵P；一行（列）的P（矩阵）倍加到另一行（列）上；可以得到如下类型的分块矩阵：

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- $\begin{bmatrix} O & I_n \\ I_m & O \end{bmatrix}$,
- $\begin{bmatrix} P & O \\ O & I_n \end{bmatrix}, \begin{bmatrix} I_m & O \\ O & P \end{bmatrix}$,
- $\begin{bmatrix} I_m & P \\ O & I_n \end{bmatrix}, \begin{bmatrix} I_m & O \\ P & I_n \end{bmatrix}$

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- 用这些矩阵左乘任一个分块矩阵

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

- 只要分块乘法能够进行，其结果就是对它做相应的变换：

- $\begin{bmatrix} O & I_m \\ I_n & I_n \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix}=\begin{bmatrix} C & D \\ A & B \end{bmatrix}$，

- $\begin{bmatrix} P & O \\ O & I_n \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix}=\begin{bmatrix} PA & PB \\ C & D \end{bmatrix}$，

- $\begin{bmatrix} I_m & O \\ P & I_n \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix}=\begin{bmatrix} A & B \\ C+PA & D+PB \end{bmatrix}$

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在$\begin{bmatrix} I_m & O \\ P & I_n \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ C+PA & D+PB \end{bmatrix}$式中，

- 适当选择P，可使 C + PA = O。例如A 可逆时，选
- P = -CA^{-1} ，则 C + PA = O，于是
- $\begin{bmatrix} I_m & O \\ P & I_n \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ C+PA & D+PB \end{bmatrix} = \begin{bmatrix} A & B \\ O & D-CA^{-1}B \end{bmatrix}$

- 这种上（下）三角型的分块矩阵在矩阵的运算中非常有用！

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思考题

- 1. 设 $T=\begin{bmatrix} A & O \\ C & D \end{bmatrix}$, A,D 可逆，求 T^{-1} .
- 2. 设 $T_1=\begin{bmatrix} A & B \\ C & D \end{bmatrix}$，且 T_1 , D 均可逆，试证
- （A - B D^{-1}C）$^{-1}$ 存在，并求T_1^{-1}.

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Assingment

- Ex2.4: 23

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§ 2.5 Matrix Factorizations

- 1. The LU Factorization（ LU分解 ）
- 2. An LU Factorization Algorithm

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§ 2.5 Matrix Factorizations

A factorization of a matrix A is an equation that expresses A as a product of two or more matrices.

一些预备知识：

lower triangular matrix :下三角矩阵

upper triangular matrix :上三角矩阵

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LU Factorization

Assume that A is m×n matrix and can be row reduced to echelon form **without row interchanges.** Then A can be written in:

$$A = LU$$

where L is **m×m** lower triangular matrix with 1's on the diagonal and U is m×n echelon form of A.

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{U}$$

Such a factorization is called an LU factorization.

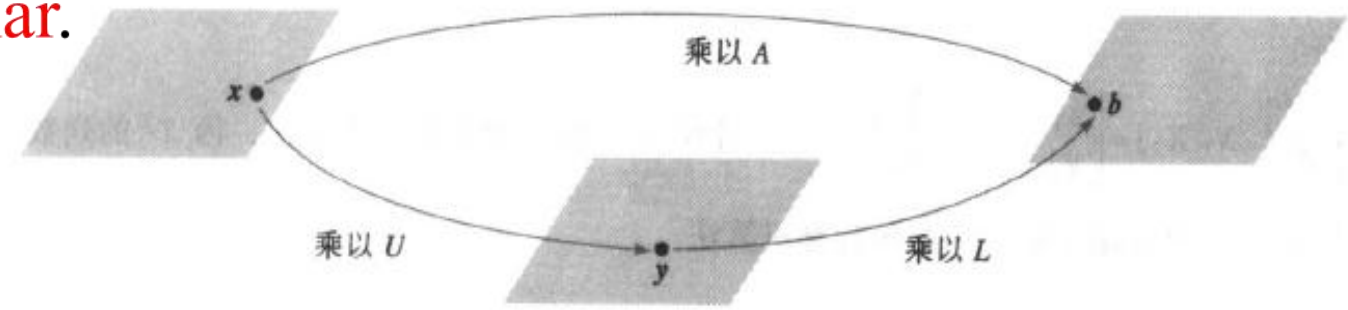
L : a **unit lower triangular matrix.**

Before studying how to construct L and U, we should look at why they are so useful?

When $A = LU$, $Ax = b$ can be written as $L(Ux) = b$. writing y for Ux, we can find x by solving the pair of equations

$$Ly = b$$
$$Ux = y$$

First solve $Ly = b$ for y, and then solve $Ux = y$ for x. Each equation is easy to solve because L and U are triangular.



- Example ： It can be verified that

$$A=\begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -5 & 1 & 0 \\ -3 & 8 & 3 & 1 \end{bmatrix}\begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix}=LU$$

Use this LU factorization of A to solve Ax = b, where

$$\boldsymbol{b}=\begin{bmatrix} -9 \\ 5 \\ 7 \\ 11 \end{bmatrix}$$

- Solution:

$$[L \quad \boldsymbol{b}] = \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ -1 & 1 & 0 & 0 & 5 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = [I \quad \boldsymbol{y}]$$

$$[U \quad \boldsymbol{y}] = \begin{bmatrix} 3 & -7 & -2 & 2 & -9 \\ 0 & -2 & -1 & 2 & -4 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}, \quad \boldsymbol{x} = \begin{bmatrix} 3 \\ 4 \\ -6 \\ -1 \end{bmatrix}$$

2. An LU Factorization Algorithm

Suppose A can be reduced to an echelon form U using only row replacement that add a multiple of one row to another row below it.

Then, there exist a series of unit lower triangular elementary matrix $E_p \cdots E_1 A = U$ (3)

Then $A = (E_p \cdots E_1)^{-1} U = LU$

Whe $L = (E_p \cdots E_1)^{-1}$ (4)

The product unit lower triangular matrices are is unit lower triangular.

Thus L is unit lower triangular .

§ 2.5 Matrix Factorizations

Algorithm for An *LU* Factorization

1. Reduce *A* to an echelon form *U* by a sequence of row replacement operations, if possible.

2. Place entries in *L* such that the same sequence of row operations reduces *L* to *I*.

说明：步骤1并不是总是可行（并不是所有矩阵可以进行LU分解）；
步骤2说得比较笼统，实际中应该如何做呢？

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- Example ： Find an LU factorization of

$$A=\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

- Solution：

Since A has four rows, L should be 4×4

$$L=\begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix}$$

- Row reduction of A to an echelon form U:

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix} = A_1$$

$$\sim A_2 = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix} = U$$

- At each pivot column, divide the highlighted entries by the pivot and place the result into L:

$$\begin{bmatrix} 2 \\ -4 \\ 2 \\ -6 \end{bmatrix} \quad \begin{bmatrix} 3 \\ -9 \\ 12 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 4 \end{bmatrix} \quad [5]$$

$$\begin{matrix} \div 2 & \div 3 & \div 2 & \div 5 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{matrix}$$

$$\begin{bmatrix} 1 & & & \\ -2 & 1 & & \\ 1 & -3 & 1 & \\ -3 & 4 & 2 & 1 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$

Assignment

- Ex2.5: 9 ,19

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2. 6, 2.7 Pass

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§ 2.8 Subspace of R^n

- Subspace of R^n
- Column Space and Null Space of a Matrix
- Basis for a Subspace

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§ 2.8 Subspace of R^n

- **Subspace（子空间）of R^n**

  Definition: A subspace of R^n is any set H in R^n that has three properties：

  a. The **zero vector** is in H.

  b. For each **u** and **v** in H, the sum **u + v** is in H.

  c. For each **u** in H and each scalar c, the vector **cu** is in H.

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- Example: If $\mathbf{v}_1$ and $\mathbf{v}_2$ are in R^n and H = span{$\mathbf{v}_1$,$\mathbf{v}_2$}, then H is a subspace of R^n .To verify this statement, note that the zero vector is in H. Now take two arbitrary vectors in H, say,

$$\boldsymbol{u} = s_1\boldsymbol{v}_1 + s_2\boldsymbol{v}_2 \ , \boldsymbol{v} = t_1\boldsymbol{v}_1 + t_2\boldsymbol{v}_2 \ ,$$

Then

$$\boldsymbol{u} + \boldsymbol{v} = (s_1 + t_1)\boldsymbol{v}_1 + (s_2 + t_2)\boldsymbol{v}_2$$

which shows that **u** + **v** is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$ and hence is in H. Also, for any scalar c, the vector cu is in H.

$$c\boldsymbol{u} = c(s_1\boldsymbol{v}_1 + s_2\boldsymbol{v}_2) = (cs_1)\boldsymbol{v}_1 + (cs_2)\boldsymbol{v}_2$$

§ 2.8 Subspace of R^n

- Example : For $v_1,\ldots,v_p$ in R^n, the set of all linear combinations of $v_1,\ldots,v_p$ is a subspace of R^n. We shall now refer to Span $\{v_1,\ldots,v_p\}$ as the subspace spanned by $v_1,\ldots,v_p$.

- R^n is a subspace of itself .
- **Zero Subspace** is the set consisting of only the zero vector in R^n .

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§ 2.8 Subspace of R^n

- Column Space (列向量空间) and

  Null Space (零空间) of a Matrix

Definition :

1) The column space of a matrix A is the set Col A of all linear combinations of the column of A.

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§ 2.8 Subspace of R^n

2) The null space(零空间) of a matrix A is the set Nul A of all solutions to the homogeneous equation $Ax = 0$.

Theorem 12

The null space of an m*n matrix A is a subspace of R^n. Equivalently, the set of all solution to a system Ax = 0 of m homogeneous linear equation in n unknowns is a subspace of R^n .

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Proof :

The zero vector is in Nul A (because $A0 = 0$).

Suppose u and v in Nul A. That is, $Au = 0$ and $Av = 0$. Then, by a property of matrix multiplication,

$$A(u+v) = Au + Av = 0 + 0 = 0$$

Thus $u + v$ satisfies $Ax = 0$, and so $u + v$ is in Nul A.

Also, for any scalar c,

$$A(cu) = c(Au) = c(0) = 0,$$

which shows that cu is in Nul A.

§ 2.8 Subspace of R^n

- **Basis（基） for a Subspace**

Definition :

A basis for a subspace H of R^n is a linearly independent set in H that spans H.

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补充

- 极大线性无关向量组；两个向量组等价；
- 定义：一个向量组的一个部分组称为一个极大线性无关组，如果这个部分组本身是线性无关的，并且从这向量组中任意添加一个向量（如果还有的话），所得的部分向量组都线性相关。

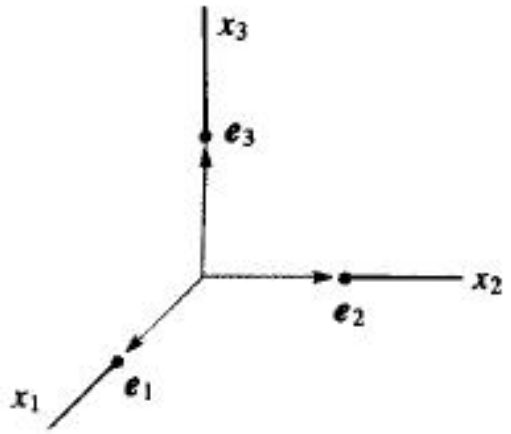
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- 定义：当向量α是向量组β_1，β_2，…，β_s的一个线性组合时，就称α可以经向量组β_1，β_2，…，β_s线性表出。
- 定义：如果两个向量组互相可以线性表出，它们就称为等价的.

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Example :The columns of an identity n×n matrix form a basis for all of R^n. Its columns are denoted by $e_1,\ldots,e_n$. The set $\{ e_1,\ldots,e_n \}$ is called the standard basis for R^n.

$$\boldsymbol{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \boldsymbol{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \cdots, \boldsymbol{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$



- Example : Find a basis for the null space of the matrix.

$$A=\begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Solution: First, write the solution of Ax = 0 in parametric vector form:

$$A \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \qquad \begin{aligned} x_1 - 2x_2 \quad - \; x_4 + 3x_5 &= 0 \\ x_3 + 2x_4 - 2x_5 &= 0 \\ 0 &= 0 \end{aligned}$$

The general solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$= x_2 u + x_4 v + x_5 w$$

are u,v,w linearly independent?

- Example : Find a basis for the column space of the matrix

$$B=\begin{bmatrix}1 & 0 & -3 & 5 & 0\\ 0 & 1 & 2 & -1 & 0\\ 0 & 0 & 0 & 0 & 1\\ 0 & 0 & 0 & 0 & 0\end{bmatrix}$$

Solution: Denote the columns of B by $b_1,\ldots,b_5$ and note that b_3 and b_4 are combinations of the pivot columns.

$$\boldsymbol{b}_3 = -3\boldsymbol{b}_1 + 2\boldsymbol{b}_2, \boldsymbol{b}_4 = 5\boldsymbol{b}_1 - \boldsymbol{b}_2$$

then, substituting for b_3 and b_4 , we can write v in the form

$$\boldsymbol{v} = c_1\boldsymbol{b}_1 + c_2\boldsymbol{b}_2 + c_3(-3\boldsymbol{b}_1 + 2\boldsymbol{b}_2) + c_4(5\boldsymbol{b}_1 - \boldsymbol{b}_2) + c_5\boldsymbol{b}_5$$

Thus, we have $\text{Span}\{b_1, b_2, b_3\}=\text{Col B}$

Furthermore, one can see that b_1, b_2, b_3 are linearly independent, and thus they form a basis for Col B

§ 2.8 Subspace of R^n

Theorem 13 ?

The pivot columns of a matrix A form a basis for the column space of A.

- **Warning:**

  Be careful to use pivot columns of A itself for the basis of Col.A. The columns of an echelon form B are often not in the column space of A.

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Assignment

- Ex2.8: 23,25

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§ 2.9 Dimension and Rank

- Coordinate Systems
- The Dimension of a Subspace
- Rank and the Invertible Matrix Theorem

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§ 2.9 Dimension and Rank

Coordinate Systems (坐标系)

Definition: Suppose the set $B = \{\mathbf{b}_1,\ldots,\mathbf{b}_p\}$ is a basis for a subspace H. For each $\mathbf{x}$ in H, the coordinates of $\mathbf{x}$ relative to the basis B (基B下的坐标x) are the weights $c_1,\ldots c_p$ such that $x= c_1\mathbf{b}_1+\ldots+c_p\mathbf{b}_p$, and the vector in R^p

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

is called the coordinate vector (坐标向量) of $\mathbf{x}$ (relative to B) or the B-coordinate vector of $\mathbf{x}$.

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- Example : Let $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}, \mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$.

Then *B* is a basis for H = Span{v_1,v_2} because v_1 and v_2 are linearly independent.

Determine if **x** is in *H*, and if it is, find the coordinate vector of **x** relative to *B*.

Solution: If **x** is in *H*, then the following vector equation is consistent:

$$c_1 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

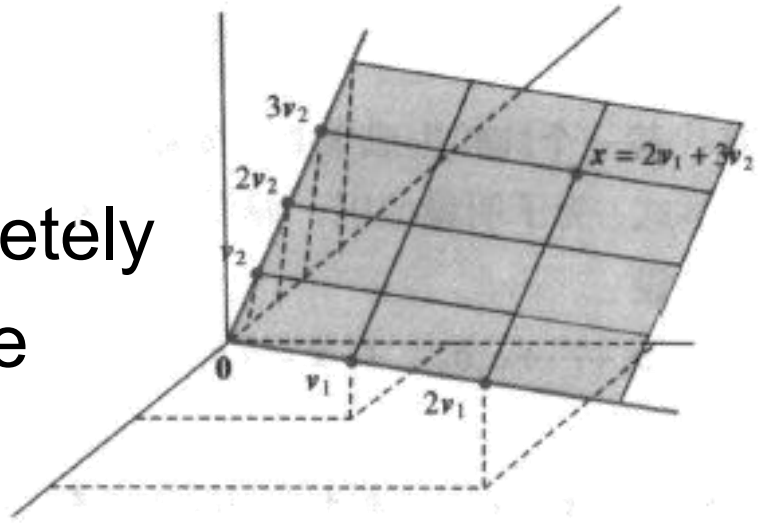
- The scalars c_1 and c_2, if they exist, are the B-coordinates of **x**.

Row operations show that

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus $c_1 = 2, c_2 = 3, [x]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. , The basis B determines a "coordinate system".

Note: Although points in H are also in R^3, they are completely determined by their coordinate vectors, which belong to R^2.



The Dimension（维数） of a Subspace

Definition :The **dimension** of a nonzero subspace *H*, denoted by dim *H*, is the number of vectors in any basis for *H*. The dimension of the zero subspace {**0**} is defined to be zero.

The space R^n has dimension n.

Every basis for R^n consists of n vectors.

A plane through 0 in R^3 is two-dimensional, and a line through 0 is one-dimensional.

- Definition : The rank (秩) of a matrix *A*, denoted by rank *A*, is the dimension of the column space of *A*.
- Example: Determine the rank of the matrix.

$$A=\begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 4 & 7 & -4 & -3 & 9 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix}$$

Solution: Reduce *A* to echelon form:

$$A \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & -6 & 4 & 14 & -20 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

主元列

The matrix *A* has 3 pivot columns, so rank A = 3.

§ 2.9 Dimension and Rank

Theorem 14 The Rank Theorem

If a matrix A has n columns, then rank A + dim Nul A = n.

Theorem 15 The Basis Theorem

Let H be a p-dimensional subspace of R^n. Any linearly independent set of exactly p elements in H is automatically a basis for H. Also, any set of p elements of H that spans H is automatically a basis for H.

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§ 2.9 Dimension and Rank

Rank and the Invertible Matrix Theorem

Theorem The Invertible Matrix Theorem (continued)

Let A be an n*n matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.

m. The columns of A form a basis of R^n

n. $\text{Col } A = \mathbb{R}^n$.

o. $\dim \text{Col } A = n$.

p. $\text{rank } A = n$.

q. $\text{Nul } A = \{\mathbf{0}\}$.

r. $\dim \text{Nul } A = 0$.

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回顾

a. A is an invertible matrix.
b. A is row equivalent to the identity matrix $\mathbf{I_n}$.
c. A has n pivot positions.
d. The equation Ax = 0 has only the trivial solution.
e. The columns of A form a linearly independent set.
f. The linear transformation $x \rightarrow Ax$ is one-to-one.
g. The equation Ax = b has at least one solution for each b in R^n.
h. The columns of A span R^n.
i. The linear transformation $x \rightarrow Ax$ maps R^n onto R^n.
j. There is an $n \times n$ matrix C such that CA = I.
k. There is an $n \times n$ matrix D such that AD = I.
l. A^T is an invertible matrix.

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Proof:

Statement (m) is logically equivalent to statements (e) and (h) regarding linear independence and spanning.

The other five statement are linked to the earlier ones of the theorem by the following chain of almost trivial implications:

(g)=>(n)=>(o)=>(p)=>(r)=>(q)=>(d)

Statement (g) which says that the equation Ax = b has at least one solution for each b in R^n, implies (n), because Col A is precisely the set of all b such that the equation Ax = b is consistent.

The implications (n)=>(o)=>(p) follow from the definitions of dimension and rank. If the rank of A is n, the number of columns of A, then dim Nul A=0, by the Rank theorem, and so Nul A = {0}. Thus (p)=>(r)=>(q).

Also, (q) implies that the equation Ax=0 has only the trivial solution, which is statement (d). Since statements (d) and (g) are already know to be equivalent to the statement that A is invertible.

Assignment

- Ex2.9: 5,13

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